

Session 11 Assignment Problems-Strings

Write the code for the following problems.

Use separate functions for input, each type of processing, and output. Avoid global variables by passing parameters and returning results. Create test data to validate the accuracy of each program. Add comments at the top of the program and include references to any resources used.

1. Create a program that asks the user for a single line of text containing a first name and last name, such as Firstname Lastname. Use string functions/methods to parse the line and print out the name in the form last name, first initial, such as Lastname, F. Include a trailing period after the first initial. Handle invalid input errors, such as extra spaces or missing name parts.

Input	Process	Output
fname + lname	Input string	Formatted name
	Strip extra spaces	Error msg if incorrect values are inputted
	Cut into parts	
	Validate 2 parts	
	Extract first and last name	
	Lname, first initial	

2. Create a program that asks the user for a line of text. Use string functions/methods to delete leading, trailing, and duplicate spaces, and then print the line of text backwards. For example:

The cat in the hat tah eht ni tac eht

Use separate subroutines/functions/methods to implement input, each type of processing, and output. Avoid global variables by passing parameters and returning results.

Input	Process	Output
Line of text (to be reversed)	Remove leading spaces, trailing, duplicate spaces	Display backwards text
	Print line of the text backwards	

3. Create a program that asks the user for a line of comma-separated-values. It could be a sequence of test scores, names, or any other values. Use string functions/methods to parse the line and print out each item on a separate line. Remove commas and any leading or trailing spaces from each item when printed.

Input	Process	Output
Line of comma-separated-values	Split string with commas	Print out each item on an individual item on individual line
	Remove commas, leading and trailing spaces	
	Loop	

4. Create a program that asks the user for a line of text. Then ask the user for the number of characters to print in each line, the number of lines to be printed, and a scroll direction, right or left. Using the given line of text, duplicate the text as needed to fill the given number of characters per line. Then print the requested number of lines, shifting the entire line's content by one character, left or right, each time the line is printed. The first or last character will be shifted / appended to the other end of the string. For example, if shifting the line to the left:

Repeat this. Repeat this.

Repeat this. Repeat this. R

peat this. Repeat this. Re

Or if shifting the line to the right:

Repeat this. Repeat this.

Repeat this. Repeat this.

. Repeat this. Repeat this

Input	Process	Output
Line of text	Repeat text to fill required line in length	Shifted lines L/R
# of characters for each line	String to match line width	
# of lines to be printed	Shift string L/R each time	
Choose Direction L/R	Wrap characteristics from one side to another	

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